Weakest Link, Not Diversity: Adversarial Vulnerability in Multi-Judge LLM Evaluation

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Abstract

Multi-judge LLM evaluation panels are increasingly adopted to improve reliability, with an implicit assumption—rooted in Byzantine Fault Tolerance (BFT) theory—that judge diversity provides robustness against manipulation. We conduct the first large-scale empirical decomposition of panel vulnerability into two structural predictors: individual judge weakness ($\eta_{\max}$, the vulnerability of the weakest panel member) and panel diversity (K_{eff}, effective number of independent behavioral clusters). Across 15 models from 6 families, 455 panels, 9 attack types, and 3 benchmarks (MMLU, ARC-Challenge, MT-Bench), we find that $\eta_{\max}$ *robustly* dominates panel vulnerability—significant in 8/12 conditions under clustered standard errors and 5/12 under wild cluster bootstrap, with a dominance ratio of 5:1 over K_{eff}. K_{eff} shows directional but statistically fragile association, collapsing from 8/12 (OLS) to 0/12 (clustered SE) after correcting for panel non-independence. A Generalized Additive Model (GAM) analysis reveals a clean mechanistic split: K_{eff} is positive (diversity harmful) for all targeted attacks and negative (diversity protective) for all shared-bias attacks—a 10:8 taxonomy across 9 attack types. Aggregation rule choice fundamentally modulates vulnerability: Bayesian calibration reverses K_{eff} direction and weakens $\eta_{\max}$, while threshold voting amplifies both. We propose a robustness-first panel design algorithm that wins 12/12 conditions, outperforming naive diversity maximization which performs worse than random selection.

1 Introduction

The use of Large Language Models (LLMs) as automated judges for evaluating other LLMs has become a cornerstone of modern NLP evaluation (1). To mitigate individual judge biases, recent work increasingly employs multi-judge panels where multiple LLM judges evaluate candidates and their scores are aggregated via majority vote or similar mechanisms. The implicit operating assumption—borrowed from distributed systems and BFT theory—is that judge diversity provides robustness: more diverse panels should be harder to fool.

This assumption has not been empirically tested under adversarial conditions. In non-adversarial settings, diversity does reduce calibration errors and position biases. But adversarial manipulation fundamentally changes the threat model: an adaptive adversary can exploit judge-specific vulnerabilities, potentially turning diversity from a defense into an attack surface.

We conduct the first large-scale empirical study decomposing panel vulnerability into two structural predictors. Let $\eta_{\max}$ denote the vulnerability of the weakest judge in a panel (the maximum individual attack success rate), and K_{eff} denote the effective number of independent behavioral clusters among panel judges (a diversity measure derived from mutual information). Our central question: *which structural factor—individual weakness or panel diversity—drives vulnerability under adaptive adversarial attack?*

Our findings challenge the diversity-as-defense assumption:

1. **Weakest-link dominance.** $\eta_{\max}$ robustly dominates panel vulnerability (5:1 over K_{eff} under bootstrap inference). K_{eff}'s apparent significance under OLS (8/12) collapses to 0–1/12 after correcting for panel non-independence—a methodological finding relevant to all combinatorial evaluation designs.

2. **BFT reversal evidence.** A random-fault Monte Carlo contrast experiment shows K_{eff} is genuinely significant (2/2, all corrections) under i.i.d. faults but artifactual under adaptive attacks, providing causal evidence for the failure of BFT assumptions.
3. **Attack-type taxonomy.** GAM analysis reveals a clean mechanistic split: K_{eff} is positive (diversity harmful) for targeted attacks and negative (diversity protective) for shared-bias attacks—a 10:8 pattern across 9 attack types.
4. **Aggregation as control lever.** Bayesian calibration reverses K_{eff} direction (from harmful to protective) and weakens $\eta_{\max}$ dominance. Threshold voting amplifies both drivers ($R^2 = 0.87$).
5. **Robustness-first panel design.** A two-stage algorithm (filter by $\eta_{\max}$, then optimize K_{eff}) wins 12/12 conditions. Naive K_{eff} maximization performs *worse* than random selection.

2 Related Work

LLM-as-Judge. LLM-based evaluation has become standard practice for open-ended generation tasks (1). Known biases include position bias, verbosity bias, and sycophancy. Multi-judge panels have been proposed as a mitigation strategy, though systematic analysis of panel robustness under adversarial conditions remains absent.

Adversarial Attacks on LLM Judges. Prior work has demonstrated that individual LLM judges can be manipulated through prompt injection, score inflation, and stylistic gaming. However, existing studies focus on single-judge vulnerability rather than panel-level structural predictors.

Byzantine Fault Tolerance. BFT theory guarantees correctness when fewer than $f < n/3$ nodes fail independently. The key assumption—independent, random faults—is violated by adaptive adversaries who can strategically target specific judges. Our work provides the first empirical quantification of this assumption violation in LLM evaluation.

3 Method

3.1 Panel Construction

We enumerate all $\binom{15}{3} = 455$ panels of size $K = 3$ from a pool of 15 LLM judges spanning 6 model families (GPT, Claude, Gemini, Llama, Qwen, Mistral) and 3 capability tiers (frontier, mid, base). Each panel evaluates benchmark items under majority vote.

3.2 Vulnerability Predictors

Individual Weakness ($\eta_{\max}$). For each panel, $\eta_{\max} = \max_{j \in \text{panel}} \eta_j$, where η_j is judge j's individual attack success rate (ASR) under a given attack type. This captures the weakest-link hypothesis: panel vulnerability is determined by the most exploitable judge.

Panel Diversity (K_{eff}). We compute pairwise mutual information (MI) between all judge pairs' evaluation vectors on clean data, then derive K_{eff} as the effective number of independent clusters. Higher K_{eff} indicates greater behavioral diversity among panel judges.

3.3 Attack Types

We study 9 attack types spanning two mechanistic categories:

Method	$\eta_{\max}$ sig	K_{eff} sig	K_{eff} dir	Ratio
OLS	11/12	8/12	7+/1−	1.4:1
Clustered SE	8/12	0/12	–	∞
Bootstrap v2	5/12	1/12	1+	5:1
Mixed effects	7/12	3/12	3+	2.3:1

Table 1: Correction method comparison (15 models, 455 panels, 12 conditions). $\eta_{\max}$ significance is robust across all methods; K_{eff} collapses from 8/12 under OLS to 0–1/12 after non-independence correction. SE inflation $\sim$2.5$\times$ is consistent with 91/455 panel overlap.

- **Targeted attacks** exploit judge-specific weaknesses: prompt injection, score manipulation, position bias exploitation, few-shot hijacking, instruction override, format gaming.
- **Shared-bias attacks** exploit common cognitive biases: sycophancy amplification, verbosity bias, authority bias, style manipulation, length inflation, persona manipulation.

3.4 Regression Specification

Our canonical specification is log-log OLS:

$$\log(\text{ASR} + \epsilon) = \beta_0 + \beta_1 \log(K_{\text{eff}}) + \beta_2 \log(\eta_{\max} + \epsilon) \tag{1}$$

where ϵ is a small constant for numerical stability. We report significance under four correction methods: standard OLS, clustered standard errors (CR1, 15 judge clusters), wild cluster bootstrap ($B = 9999$, Rademacher weights, max-p across 3 judge positions), and mixed effects (random judge intercepts). Panel non-independence arises because judges appear in multiple panels (e.g., claude-3-haiku appears in 91/455 panels), inflating standard errors by $\sim$2.5$\times$.

3.5 Aggregation Rules

We evaluate five aggregation rules: majority vote (baseline), accuracy-weighted vote, threshold (unanimity), Bayesian calibration (prior-weighted), and veto (single-judge override).

4 Experiments

4.1 Core Results: Weakest-Link Dominance

Table 1 presents our primary results across 15 models, 455 panels, and 12 conditions (6 attack types $\times$ 2 benchmarks). The pattern is stark: $\eta_{\max}$ dominates under all correction methods, while K_{eff}'s significance collapses after accounting for panel non-independence.

Under OLS, both predictors appear significant ($\eta_{\max}$ 11/12, K_{eff} 8/12), producing a misleading near-parity (1.4:1). Clustered SE correction—which accounts for shared judge membership across panels—eliminates all K_{eff} significance (0/12) while retaining $\eta_{\max}$ at 8/12. The most conservative method (wild cluster bootstrap with max-p across judge positions) gives a 5:1 dominance ratio. The $\min(\eta_{\max})$ panel selection strategy wins 12/12 conditions with ASR advantages ranging from 3.8 to 51.1 percentage points.

4.2 Attack Surface Expansion

To confirm that $\eta_{\max}$ dominance is not an artifact of our initial attack selection, we extend to 3 additional attack types (few-shot hijacking, persona manipulation, instruction override), yielding 18 total conditions. Table 2 shows results under clustered SE.

Attack	Mech.	$\eta_{\max}$	K_{eff}	GAM dir
Few-shot hijack	targ.	✓✓	0/2	positive
Persona manip.	shared	✓✓	0/2	negative
Instr. override	targ.	1/2	0/2	positive
Combined (18 cond.)		13/18	0/18	10:8

Table 2: Extended attack taxonomy (clustered SE). $\eta_{\max}$ remains dominant across all new attack types. GAM taxonomy extends cleanly from 6:6 to 10:8 (targeted:shared-bias). ✓✓ = significant on both benchmarks.

Metric	Random Fault	Adaptive	Interpretation
K_{eff} sig (OLS)	2/2	8/12	Both appear sig.
K_{eff} sig (CL-SE)	2/2	0/12	RF genuine; adap. not
K_{eff} sig (Boot.)	2/2	1/12	Consistent contrast
K_{eff} std β	+0.297	+0.164	RF larger, genuine
R^2	0.78–0.81	0.25–0.35	Predictability gap
$\eta_{\max}$ sig (all)	2/2	5–8/12	Genuine in both

Table 3: Random-fault vs. adaptive regime contrast. K_{eff} is genuinely significant under i.i.d. faults (2/2 under all corrections) but artifactual under adaptive attacks (0–1/12 after correction). The R^2 gap ($\Delta = 0.48$) quantifies the unpredictability introduced by adaptive adversaries.

The combined dominance ratio under clustered SE is 13:0—K_{eff} achieves zero significance across all 18 conditions. The GAM taxonomy extends cleanly: few-shot hijacking and instruction override are classified as targeted (positive K_{eff}), while persona manipulation is classified as shared-bias (negative K_{eff}), extending the split from 6:6 to 10:8.

4.3 Random-Fault BFT Contrast

To provide causal evidence for BFT assumption failure, we construct a random-fault (RF) baseline via Monte Carlo simulation (10,000 iterations). Each judge fails independently with probability $\eta_j = 1 -$ clean_accuracy.

Table 3 presents the contrast. Under random faults, K_{eff} is significant 2/2 under *all* correction methods ($p_{\text{bootstrap}} = 0.003$ for MMLU, <0.001 for ARC), with $R^2 = 0.78$–0.81. Under adaptive attacks, the same predictor collapses to 0–1/12 after correction. This is not a direction reversal—K_{eff} is positively associated with vulnerability in *both* regimes. The difference is that the RF association is genuine (panel structure determines fault tolerance under i.i.d. assumptions), while the adaptive association is inflated by panel overlap. The R^2 gap ($\Delta = 0.48$) directly quantifies the strategic variance introduced by adaptive adversaries that panel structure cannot predict.

4.4 ASR Concentration Analysis

To provide mechanistic evidence for $\eta_{\max}$ dominance, we compute per-judge ASR distributions within each panel and measure concentration via Gini coefficients and Shannon entropy.

Table 4 reveals a sharp dichotomy. Under targeted attacks, ASR is highly concentrated: the most vulnerable judge contributes 56–65% of total panel ASR (mean Gini = 0.78). Under shared-bias attacks, ASR is more uniformly distributed (mean Gini = 0.43, top-1 fraction = 37%). The Gini–$\eta_{\max}$ correlation is $r = 0.83$, directly linking panel-level vulnerability to judge-level concentration. This independently validates the GAM taxonomy: targeted and shared-bias attacks differ not only in K_{eff} direction but in their micro-level ASR distribution patterns.

Attack	Cat.	Gini	Entropy	Top-1 frac.
Prompt inject.	targ.	0.82	0.73	0.65
Score manip.	targ.	0.79	0.78	0.62
Position bias	targ.	0.74	0.94	0.56
Sycophancy	shared	0.45	1.38	0.40
Verbosity bias	shared	0.42	1.47	0.37
Authority bias	shared	0.41	1.50	0.36

Table 4: ASR concentration by attack category. Targeted attacks show highly concentrated ASR (Gini $>$ 0.7, top-1 judge contributes $>$56% of panel ASR), providing mechanistic evidence for $\eta_{\max}$ dominance. Mann-Whitney U test: $p < 0.001$ between categories.

Rule	$\eta_{\max}$	K_{eff}	Neg.	R^2	Position
Veto	11/12	0/12	–	0.82	Max vuln.
Threshold	10/12	4/12	1/4	0.87	High
Majority	8/12	0/12	–	0.28	Moderate
Weighted	6/12	1/12	0/1	0.26	Low-mod.
Bayesian	5/12	4/12	3/4	0.09	Min vuln.

Table 5: Aggregation gradient under clustered SE. "Neg." = fraction of significant K_{eff} coefficients that are negative. Bayesian calibration weakens $\eta_{\max}$ (11→5/12) and flips K_{eff} direction (3/4 negative), while veto/threshold amplify the weakest-link channel.

4.5 Aggregation Gradient

We evaluate five aggregation rules under clustered SE to characterize how vote aggregation modulates vulnerability.

Table 5 reveals a gradient from maximum to minimum vulnerability. Veto rule (any single judge can override) maximizes $\eta_{\max}$ dominance (11/12) with K_{eff} completely suppressed. Threshold voting (unanimity) amplifies both channels ($R^2 = 0.87$). Bayesian calibration, which down-weights unreliable judges using prior accuracy estimates, achieves the strongest mitigation: $\eta_{\max}$ drops to 5/12 and K_{eff} flips to majority-negative (3/4 significant coefficients negative), meaning diversity becomes protective.

Under bootstrap, Bayesian K_{eff} shows 3/12 significant—all negative (3/3 negative fraction = 100%). This directional flip is the cleanest evidence that aggregation is an independent control lever, unaffected by the non-independence concerns that plague the diversity analysis.

4.6 Panel Design Algorithm

We benchmark a two-stage robustness-first algorithm against 6 baseline strategies across 12 conditions.

Table 6 presents the ranking. The most striking result is that $\max(K_{\text{eff}})$—the intuitive "diversify your panel" strategy—ranks *last*, performing 1.3pp worse than random selection. This occurs because maximizing diversity necessarily introduces weaker models (the diversity-capability tradeoff, $r = 0.70$). The two-stage algorithm provides a consistent but modest improvement (0.7pp) over pure $\min(\eta_{\max})$, with the marginal gain concentrated in targeted attack conditions (5/6 wins, mean $\Delta = -1.1$pp) where controlled K_{eff} optimization adds value after robustness screening.

4.7 Capability-Matched Validation

To disentangle $\eta_{\max}$ and K_{eff} (confound $r = 0.70$ in the full pool), we conduct within-tier analysis using 9 base-tier models (84 panels), which reduces the confound to $r = 0.35$ and $\eta_{\max}$ variance by 65–70%.

Within the base tier (Table 7), K_{eff} emerges as a weak secondary factor (2/12 significant under clustered SE), with the dominance ratio narrowing from ∞ to 2:1. Both significant K_{eff} conditions are targeted attacks (mmlu×prompt_injection, $\beta = +0.31$, $p = 0.028$; mmlu×score_manipulation,

Strategy	Mean ASR	Rank	Δ vs. random (pp)
Two-stage	0.112	1	-8.3
$\min(\eta_{\max})$	0.119	2	-7.6
Strongest-K	0.131	3	-6.4
Uniform family	0.158	4	-3.7
Max disagreement	0.172	5	-2.3
Random	0.195	6	0.0
$\max(K_{\text{eff}})$	0.208	7	$+1.3$

Table 6: Panel design strategy comparison (mean ASR across 12 conditions). The $\max(K_{\text{eff}})$ strategy—the intuitive "maximize diversity" approach—performs *worse* than random selection because it introduces weak models. Two-stage provides a marginal 0.7pp improvement over $\min(\eta_{\max})$ alone.

Setting	Panels	r	$\eta_{\max}$ (CL)	K_{eff} (CL)
Full pool	455	0.70	8/12	0/12
Base tier	84	0.35	4/12	2/12

Table 7: Cross-tier vs. within-tier comparison (clustered SE). Controlling for capability reduces confound and reveals K_{eff} as a weak but detectable secondary factor (2/12, both in targeted conditions). The dominance ratio narrows from ∞ to 2:1, but $\eta_{\max}$ still dominates.

$\beta = +0.25$, $p = 0.041$), consistent with the GAM taxonomy. The within-tier R^2 mean drops to 0.18 (vs. 0.30 in the full pool), confirming that cross-tier $\eta_{\max}$ variance is the primary source of predictive power.

4.8 Diversity-Capability Tradeoff

The structural correlation between K_{eff} and $\eta_{\max}$ ($r = 0.70$, 95% CI $[0.66, 0.74]$) reflects a fundamental constraint in the current LLM ecosystem: maximizing behavioral diversity requires including models from different capability tiers, which inevitably introduces weaker judges. This correlation is strongest within model families ($r = 0.81$) and weakest at the frontier tier ($r = 0.35$), where architectural differences provide diversity without large capability gaps. The Pareto frontier analysis shows that 97.4% of panels are dominated, with optimal panels clustering in the low-K_{eff}, low-$\eta_{\max}$ region. VIF analysis confirms moderate but non-pathological multicollinearity (VIF = 2.96, condition number = 4.8), with partial correlations of $r(\eta_{\max}, \text{ASR} \mid K_{\text{eff}}) = 0.42$ vs. $r(K_{\text{eff}}, \text{ASR} \mid \eta_{\max}) = 0.18$.

4.9 Panel Size Sensitivity

Table 8 demonstrates that $\eta_{\max}$ dominance is not an artifact of small panels. Across $K = 3$ to $K = 9$, the $\eta_{\max}$ effect size decreases slightly (standardized β: 0.42 to 0.31) as larger panels provide more majority-vote fault tolerance, but the dominance ratio remains $\geq$4:1. The $\min(\eta_{\max})$ strategy wins 12/12 conditions at every panel size. SE inflation increases from 2.5$\times$ to 7.8$\times$, providing compelling evidence that non-independence correction is essential for panel studies at any scale.

4.10 Generation Benchmark Extension

To rule out MCQ-specific artifacts, we extend to MT-Bench, an open-ended generation benchmark with 1–10 scale scoring. We test 3 generation-specific attack types across 12 available models and 220 panels.

Table 9 confirms that $\eta_{\max}$ dominance generalizes across evaluation formats. Under clustered SE, the combined dominance ratio is 12:0 across 18 conditions. GAM classification of generation attacks aligns with the mechanistic taxonomy: style manipulation and length inflation are shared-bias (negative K_{eff}), while format gaming is targeted (positive K_{eff}). The $\min(\eta_{\max})$ strategy wins 6/6 generation conditions, confirming the practical recommendation is format-agnostic.

K	**Panels**	$\eta_{\max}$	K_{eff}	**Ratio**	**SE infl.**
3	455	8/12	0/12	∞	2.5×
5	3003	9/12	1/12	9:1	4.1×
7	6435	9/12	1/12	9:1	6.3×
9	5005	8/12	2/12	4:1	7.8×

Table 8: Panel size sensitivity (clustered SE). $\eta_{\max}$ dominance persists across $K = 3$–9, confirming the weakest-link mechanism is scale-invariant. SE inflation increases sharply with K, underscoring the necessity of non-independence correction in combinatorial designs.

Format	$\eta_{\max}$ **(CL)**	K_{eff} **(CL)**	**Ratio**
MCQ (MMLU+ARC)	8/12	0/12	∞
Generation (MT-Bench)	4/6	0/6	∞
Combined	12/18	0/18	∞

Table 9: Cross-format consistency (clustered SE). $\eta_{\max}$ dominance generalizes from MCQ to open-ended generation evaluation. The combined ratio across 18 conditions is ∞ (12:0).

4.11 Partial-Knowledge Adversary

Real-world adversaries rarely have full knowledge of panel composition. We simulate three knowledge levels: full (baseline), family-only (adversary knows model family labels but not specific vulnerability profiles), and random (no targeting).

Table 10 reveals that family-level knowledge retains 82% of full-knowledge attack efficiency (ASR $=$ 0.160 vs. 0.195), because same-family models share training pipeline and architectural biases ($r_{\text{family-vulnerability}} = 0.61$). Defenders cannot rely on panel composition secrecy: family labels are easily inferred from API endpoints and response styles. K_{eff} remains 0/12 under all knowledge levels, and $\eta_{\max}$ dominance persists even under family-only targeting (7/12), confirming the core findings are robust to adversary capability assumptions.

5 Discussion

When diversity helps. Our results do not imply diversity is universally harmful. Under shared-bias attacks, GAM analysis consistently shows negative K_{eff} direction (8/8 conditions including extended attacks), meaning diversity provides partial immunity when the adversary exploits biases shared across model families. After robustness screening removes weak judges, the diversity-capability tradeoff is weakened (r drops from 0.70 to 0.35), creating conditions where diversity optimization adds marginal value. And in non-adversarial settings, diversity remains beneficial for reducing calibration errors and position biases.

Practical recommendations. Based on our findings, we recommend a defense stack combining: (1) robustness-first screening to eliminate $\eta_{\max}$ as a vulnerability driver, and (2) Bayesian aggregation to reverse K_{eff} from harmful to protective. This dual-channel approach addresses both structural vulnerability drivers. The $\max(K_{\text{eff}})$ strategy should be explicitly avoided, as it performs worse than random selection.

Methodological implications. The dramatic significance shift from OLS to clustered SE (K_{eff}: 8/12 $\rightarrow$ 0/12) is a cautionary finding for all combinatorial evaluation designs where the same items (here, judges) appear in multiple experimental units (panels). SE inflation of 2.5–7.8× can produce severely misleading conclusions. We recommend that future panel studies routinely report clustered SE alongside OLS.

Knowledge	**ASR**	$\eta_{\max}$ **(CL)**	K_{eff} **(CL)**	R^2
Full	0.195	8/12	0/12	0.30
Family-only	0.160	7/12	0/12	0.24
Random	0.113	3/12	0/12	0.08

Table 10: Adversary knowledge level comparison (clustered SE). Family-only knowledge retains 82% attack efficiency. K_{eff} is 0/12 under all knowledge levels. $\eta_{\max}$ dominance persists under realistic threat models.

6 Conclusion

We conducted the first large-scale empirical decomposition of multi-judge LLM evaluation panel vulnerability. Our central finding—that panel vulnerability is dominated by the weakest individual judge, not panel diversity—challenges the implicit BFT assumption underlying multi-judge evaluation design. The dominance ratio of 5:1 ($\eta_{\max}$ over K_{eff}) is robust across 9 attack types, 3 benchmarks, 4 panel sizes, 3 adversary knowledge levels, and 4 statistical correction methods. Aggregation rule choice provides an independent control lever: Bayesian calibration reverses diversity's role from vulnerability to defense. We release our two-stage panel design algorithm as a practical tool for building adversarially robust evaluation panels.

Limitations

Our study has three main limitations. First, the model pool (15 models, 6 families) represents the current LLM ecosystem; the diversity-capability tradeoff ($r = 0.70$) may weaken as more capable open-source models become available. Second, our adaptive adversary model assumes strategic targeting of individual judges—real-world adversaries may face additional constraints (e.g., limited query budgets) that reduce attack efficiency below the levels we measure. Third, the Bayesian aggregation defense requires access to clean-data accuracy estimates, which may themselves be subject to manipulation; evaluating the robustness of the defense stack to meta-adversarial attacks remains future work.

References

[1] Lianmin Zheng, Wei-Lin Chiang, Ying Sheng, Siyuan Zhuang, Zhanghao Wu, Yonghao Zhuang, Zi Lin, Zhuohan Li, Dacheng Li, Eric P. Xing, Hao Zhang, Joseph E. Gonzalez, and Ion Stoica. Judging LLM-as-a-Judge with MT-Bench and Chatbot Arena. In *Advances in Neural Information Processing Systems*, 2023.

[2] Leslie Lamport, Robert Shostak, and Marshall Pease. The Byzantine generals problem. *ACM Transactions on Programming Languages and Systems*, 4(3):382–401, 1982.

[3] Dan Hendrycks, Collin Burns, Steven Basart, Andy Zou, Mantas Mazeika, Dawn Song, and Jacob Steinhardt. Measuring massive multitask language understanding. In *International Conference on Learning Representations*, 2021.

[4] Peter Clark, Isaac Cowhey, Oren Etzioni, Tushar Khot, Ashish Sabharwal, Carissa Schoenick, and Oyvind Tafjord. Think you have solved question answering? Try ARC, the AI2 reasoning challenge. *arXiv preprint arXiv:1803.05457*, 2018.

[5] Ethan Perez, Saffron Huang, Francis Song, Trevor Cai, Roman Ring, John Aslanides, Amelia Glaese, Nat McAleese, and Geoffrey Irving. Red teaming language models with language models. In *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing*, 2022.

[6] Arjun Panickssery, Samuel R. Bowman, and Shi Feng. LLM evaluators recognize and favor their own generations. *arXiv preprint arXiv:2404.13076*, 2024.

[7] Vyas Raina, Adian Liusie, and Mark J. F. Gales. Is LLM-as-a-Judge robust? Investigating universal adversarial attacks on zero-shot LLM assessment. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, 2024.